



UNIVERSITY OF FRONTIER TECHNOLOGY, BANGLADESH
Department of Cyber Security Engineering

Project Proposal

Project Name : Sentinel-Red - An Automated TTP Emulation Engine

Course title : Object Oriented Programming Language Sessional

Course code : PROG 112

Submitted by : Name : Milton Roy (2404001) Tamim Sorder (2404002) Level : 01 Term : 02 Department of Cyber Security Engineering, UFTB	Submitted to: Md Masud Rana Lecturer, Department of Cyber Security Engineering, UFTB
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Sentinel-Red - An Automated TTP Emulation Engine

Introduction

In the contemporary cybersecurity landscape, organizations face persistent and evolving threats from sophisticated adversaries. Traditional defensive measures, while crucial, often struggle to keep pace with dynamic attack methodologies. This project, **Sentinel-Red: An Automated TTP Emulation Engine**, addresses this challenge by developing a robust, object-oriented framework designed to systematically emulate real-world adversary Tactics, Techniques, and Procedures (TTPs) against target systems. By doing so, Sentinel-Red will enable cybersecurity engineers to proactively validate the effectiveness of existing security controls, identify detection gaps, and enhance overall organizational resilience against cyberattacks. This initiative explores the engineering aspects of offensive security automation, bridging the gap between theoretical threat intelligence and practical defensive validation.

Objectives

- **Automated TTP Emulation:** To design and implement an object-oriented framework capable of executing a wide range of MITRE ATT&CK-aligned TTPs in a controlled environment.
- **Defensive Control Validation:** To provide a mechanism for systematically testing the detection and prevention capabilities of security solutions (e.g., EDR, SIEM) against emulated adversary behaviors.
- **Gap Identification:** To identify specific weaknesses or blind spots in an organization's security posture by analyzing the outcomes of TTP emulation exercises.
- **Reporting & Analysis:** To generate comprehensive, human-readable reports that detail emulation results, including detection status, and map findings back to the MITRE ATT&CK framework for actionable insights.
- **Engineering Best Practices:** To demonstrate proficiency in advanced Object-Oriented Programming (OOP) principles and software engineering design patterns (e.g., Factory, Command, Strategy, Visitor) in the context of cybersecurity tool development.

Scope

- **TTP Playbook Definition:** Creation of a flexible YAML-based schema for defining custom attack playbooks, allowing for the sequencing and parameterization of individual TTPs.
- **Cross-Platform Technique Execution:** Development of core Technique classes capable of executing commands and actions on both Windows and Linux target virtual machines.
- **Remote Execution Capabilities:** Implementation of Executor modules to facilitate secure remote execution of TTPs on isolated target systems via protocols like SSH.
- **Results Collection & Analysis:** Mechanisms for capturing the output and status of executed techniques, and for determining whether a technique was detected by simulated security controls.
- **Comprehensive Reporting:** Generation of detailed HTML reports summarizing emulation campaigns, including per-technique results, overall detection rates, and MITRE ATT&CK mapping.
- **Isolated Lab Environment:** The entire framework will operate within a virtualized, isolated lab environment to ensure safety and prevent unintended impact on production systems.

Methodology & Key Features

Sentinel-Red will be developed using a modular, object-oriented approach, emphasizing extensibility and maintainability. The core technical modules and features include:

- **TTP Playbook Engine:** A central component responsible for parsing YAML-defined playbooks and orchestrating the execution of Technique objects. This engine will support sequential and conditional execution flows, allowing for complex attack scenarios.
- **Polymorphic Technique Library:** An extensible library of BaseTechnique derived classes (e.g., ShellTechnique, PowerShellTechnique, PythonTechnique) that encapsulate specific MITRE ATT&CK TTPs. This allows for easy addition of new attack methods without altering the core framework.
- **Remote Execution Executors:** Specialized Executor classes (e.g., LocalExecutor, RemoteSSHExecutor) that abstract the method of interacting with target systems, enabling the framework to operate against diverse environments.
- **Result Aggregation & Reporting:** A robust system for collecting execution outcomes, including command output, error messages, and simulated detection status. A Reporter module will then transform this data into comprehensive, actionable reports, potentially leveraging the **Visitor Pattern** for flexible report generation.
- **Cleanup Mechanism:** Each Technique will include a cleanup() method to revert changes made during emulation, ensuring the target environment can be reset for subsequent tests.
- **MITRE ATT&CK Integration:** Direct mapping of each emulated technique to its corresponding MITRE ATT&CK ID, providing context and alignment with industry-standard threat intelligence.

Tools and Technologies

To achieve the project objectives, Sentinel-Red will leverage the following tools and technologies, with a primary focus on either C++ or Java as the development language:

Category	C++ Implementation	Java Implementation	Purpose
Development Language	C++ (C++17/20)	Java (JDK 11+)	Core programming language for the framework.
Build System	<u>CMake</u>	<u>Maven</u> or <u>Gradle</u>	Project management, dependency resolution, and compilation.
YAML Parsing	<u>yaml-cpp</u>	<u>SnakeYAML</u> or <u>Jackson Dataformat YAML</u>	Parsing attack playbooks.
Local Command Execution	<u>popen()</u> or <u>Boost.Process</u>	<u>ProcessBuilder</u>	Executing shell commands on the local machine.
Remote Command Execution	<u>libssh2</u>	<u>JSch</u> or <u>Apache Mina SSHD</u>	Securely executing commands on remote Linux targets via SSH.
Windows-Specific Interaction	Windows API calls (e.g., <u>CreateProcess</u> , <u>RegOpenKeyEx</u>)	<u>PowerShell</u> via <u>ProcessBuilder</u> (or <u>WinRM</u> client library)	Interacting with Windows OS features (e.g., Registry, WMI).
JSON Processing	<u>nlohmann/json</u>	<u>Jackson</u> or <u>Gson</u>	Parsing MITRE ATT&CK data and generating JSON reports.
Reporting/Templating	Custom HTML generation or <u>inja</u> (Jinja-like for C++)	<u>Jinja2</u> (via Python for HTML) or <u>Thymeleaf</u>	Generating interactive HTML reports.
Virtualization	VirtualBox or VMware Workstation	VirtualBox or VMware Workstation	Creating isolated lab environments (Controller & Target VMs).
Target Operating Systems	Windows 10/11, Ubuntu Server	Windows 10/11, Ubuntu Server	The victim machines for TTP emulation.
Threat Intelligence	MITRE ATT&CK STIX data, Atomic Red Team	MITRE ATT&CK STIX data, Atomic Red Team	Reference for TTPs and command examples.

Implementation Timeline

Week	Task Description	Assigned To
Week 1-2	Core Framework Setup: Define YAML playbook schema, implement <u>BaseTechnique</u> (abstract), <u>Playbook</u> (YAML parsing, sequential execution), <u>LocalExecutor</u> , and basic <u>JSONReporter</u> .	Both Members
Week 3-4	Technique Development & Remote Execution: Implement <u>ShellTechnique</u> , <u>PowerShellTechnique</u> (if applicable), and <u>RemoteSSExecutor</u> . Focus on robust error handling and <u>cleanup()</u> methods.	Md. Tamim Sorder
Week 5-6	Advanced Playbook Logic & MITRE Integration: Implement conditional execution in <u>Playbook</u> . Integrate MITRE ATT&CK data for technique mapping. Develop comprehensive <u>HTMLReporter</u> .	Milton Roy
Week 7-8	Testing, Refinement & Documentation: Conduct extensive testing of various TTPs on target VMs. Refine code, improve error messages, and finalize all project documentation (README, UML diagrams, technical report).	Both Members

Expected Outcomes

- A fully functional **Sentinel-Red** framework, implemented in either C++ or Java, capable of executing automated TTP emulation campaigns.
- A library of Technique modules covering a diverse set of MITRE ATT&CK TTPs.
- A robust Playbook engine supporting complex attack scenarios and conditional logic.
- Comprehensive, interactive HTML reports detailing emulation results, detection status, and MITRE ATT&CK mappings.
- Well-documented source code, adhering to OOP best practices and design patterns.
- A detailed technical report outlining the project's design, implementation, challenges, and findings.
- A demonstration of practical cybersecurity engineering skills in offensive security automation and defensive validation.

Challenges and Solutions

Challenge	Solution
Ensuring safe execution of TTPs without causing actual system damage.	Strict adherence to an isolated virtualized lab environment (VirtualBox/VMware) for all emulation activities. Implementing robust <u>cleanup()</u> methods for each <u>Technique</u> to revert changes.
Handling cross-platform differences (Windows vs. Linux) for command execution and system interaction.	Utilizing platform-specific libraries (e.g., Windows API for C++, <u>ProcessBuilder</u> for Java) and abstracting OS-specific logic within <u>Technique</u> and <u>Executor</u> classes through polymorphism and the Adapter Pattern .
Accurately determining if an emulated TTP was detected by target security controls.	Implementing mechanisms to monitor target VM logs (e.g., Sysmon, Event Viewer, auditd) and correlating them with executed TTPs. This may involve manual verification initially, with potential for automated log analysis in advanced stages.
Designing a flexible and extensible framework that can easily incorporate new TTPs.	Leveraging core OOP principles like inheritance and polymorphism for <u>Technique</u> classes, and design patterns like the Factory Pattern for dynamic technique instantiation from playbooks.

Conclusion

Sentinel-Red: An Automated TTP Emulation Engine represents a significant and highly relevant project for a Cybersecurity Engineering curriculum. It moves beyond theoretical understanding by providing a practical, hands-on platform for validating security defenses against real-world adversary behaviors. By meticulously designing and implementing this framework using advanced Object-Oriented Programming principles in either C++ or Java, the project will demonstrate a strong command of software architecture, offensive security concepts, and defensive validation methodologies. The expected outcomes, including a functional emulation engine and comprehensive reporting, will not only contribute to a deeper understanding of modern cyber threats but also equip the developers with invaluable engineering skills critical for a career in cybersecurity. This project aims to prove the necessity of continuous, automated security validation in maintaining a robust and resilient cyber defense posture.

References

- 1 ISO/IEC 27001:2022: Information security, cybersecurity and privacy protection — Information security management systems — Requirements.
- 2 NIST Special Publication 800-30: Guide for Conducting Risk Assessments.
- 3 MITRE ATT&CK®: A globally-accessible knowledge base of adversary tactics and techniques based on real-world observations. [<https://attack.mitre.org/>]

- 4 Atomic Red Team: A library of simple, automatable tests that replicate MITRE ATT&CK techniques. [<https://atomicredteam.io/>]
- 5 VirtualBox: Free and open-source cross-platform virtualization software. [<https://www.virtualbox.org/>]
- 6 JSch: A pure Java implementation of SSH2. [<http://www.jcraft.com/jsch/>]
- 7 yaml-cpp: A YAML parser and emitter in C++ designed for speed and efficiency. [<https://github.com/jbeder/yaml-cpp>]
- 8 ProcessBuilder (Java Documentation): Java API for creating operating system processes. [<https://docs.oracle.com/javase/8/docs/api/java/lang/ProcessBuilder.html>]
- 9 libssh2: A client-side C library implementing the SSH2 protocol. [<https://libssh2.org/>]